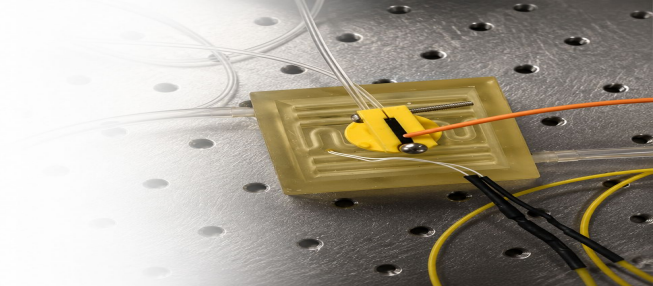


# MRI-Compatible Incubation Device for Brain Organoid Imaging

Designed and validated a temperature-controlled incubation system that preserves organoid viability during extended MRI imaging experiments



## THE CHALLENGE

High-field MRI provides powerful insights into brain organoid function but current systems cannot maintain organoid viability during long imaging sessions due to lack of temperature control and nutrient delivery

## OUR SOLUTION



### TEMPERATURE CONTROL

Maintains ~ 37°C with water bath system



### MEDIA PERFUSION

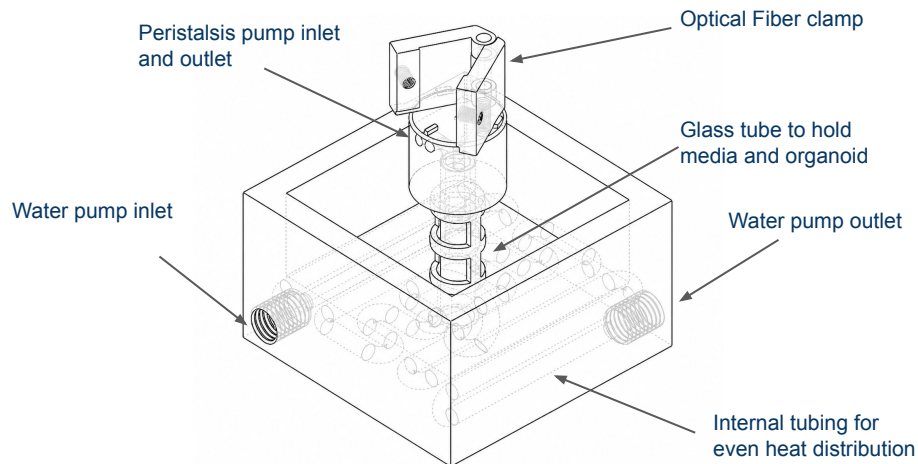
Continuous flow for nutrient delivery and waste removal



### OPTICAL FIBER PORT

Integrated fiber clamp for optogenetic stimulation

## CAD MODEL



## KEY RESULTS



### 37.3 ± 0.5°C

Stable temperature maintained inside the MR tube



### Continuous Perfusion

~400 µL volume maintained with peristaltic pump flow



### MRI-Compatible Design

Constructed with non-magnetic materials and optimized for MRI environment



### Sustained Metabolic Activity

Metabolic assay shows organoids remain viable after 5 hours of incubation



### Preserved Neuronal Activity

Calcium imaging confirms functional neural signaling post-incubation

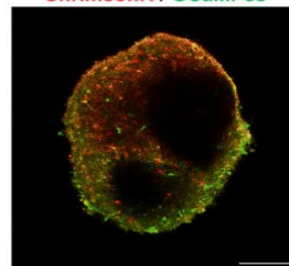


### Optogenetic compatibility

Optical fiber integrones enables precise stimulation during imaging

## CALCIUM IMAGING

ChrimsonR / GCaMP8s



- CrimsonR ⇒ Red shifted opsin
- GCaMP8s ⇒ Calcium Indicator

Brain organoid expressing ChrimsonR and GCaMP8s. Optogenetic stimulation elicited calcium responses, indicating maintained neuronal function following incubation within the MRI-compatible chamber.